



BIO 005 · HUMAN PHYSIOLOGY · YUBA COLLEGE

Week 2 Packet.

The Cell · Transport · Signaling

1 The Cell and Its Tissues
STRUCTURE

2 Transport Across the Membrane
CELLULAR PHYSIOLOGY

3 How Cells Talk to Each Other
CELL SIGNALING

HOW TO USE THIS

Read it once, then close it.

Three parts, in the order the week teaches them. Part 1 is the structure, Part 2 is what crosses it, Part 3 is how cells send messages. Read a part, then close this and put what you can into the boxes on your note sheet. What you cannot recall is the gap, and the gap is what the videos are for.

Part 1, 2 and 3 are this packet's own parts, not Silverthorn's chapters. In the book, the compartments and the membrane are Chapter 3, everything that crosses the membrane is Chapter 5, and the signaling is Chapter 6.

WHAT IS IN EACH PART, AND WHAT IT IS FOR

Element	What it is	How to use it
By the end	The competencies that part covers	Read them first. They are what the exam asks
Tables	Comparisons, side by side	Cover a column and recite it from memory
Sequences	A numbered process, step by step	Redraw the chain on blank paper without looking
Concept maps	A question with its branching answers	The question at the top is the one to ask yourself
Hold onto this	The idea the section turns on	If you remember one thing per section, this is it
Worked problems	A question, then the work in steps	Try it before you read the work. Getting it wrong first is the point

TWO COLORS

Work through the first pass in one color, before the videos. Then switch colors and add what the videos gave you that the reading did not. The gap between the two colors is the most useful thing on the page, so do not erase the first pass.

PART 1

The Cell and Its Tissues

STRUCTURE

Where the body keeps its water, what the membrane is built from, what is inside a cell, and the junctions that turn loose cells into a sheet that can do a job.

- BY THE END
1. State the volumes of the body fluid compartments in a 70 kg adult and name the dominant solute of each.
 2. Describe the fluid mosaic membrane and say what each component contributes.
 3. Name the major organelles and the three cytoskeletal filaments, and say what each one does.
 4. Tell the junction types apart by the job each one does, and predict what fails when one fails.
 5. Explain why an epithelium is polarized and why that matters for everything in Part 2.

One body, many rooms

- Every cell sits in fluid, and the composition of that fluid is what every control loop from Week 1 was defending.
- Cells only ever feel the **interstitial fluid** around them. When a later week names a regulated variable, it is almost always an extracellular value.
- Blood is the only compartment we can sample easily, and it is the smallest.

BODY FLUID COMPARTMENTS IN A 70 KG ADULT

Compartment	Fraction	Volume	What it is
Total body water	About 60 percent of body mass	42 L	Everything below, added up
Intracellular fluid	Two thirds of total body water	28 L	Inside cells. Most of your water is here
Extracellular fluid	One third of total body water	14 L	Everything outside cells
Interstitial fluid	About three quarters of the ECF	10 to 11 L	Between cells. What a cell actually touches
Plasma	About one quarter of the ECF	3 L	The fluid part of blood. About 5 L of blood in total

WHAT EACH COMPARTMENT IS MADE OF

Ion	Inside cells	Outside cells	Held that way by
Sodium	Low	High, about 140 mmol/L	The sodium potassium pump
Potassium	High	Low, about 4 mmol/L	The sodium potassium pump
Calcium	Very low	Higher	Calcium pumps
Protein	High	Low in interstitial fluid, higher in plasma	Size. Protein cannot cross freely

WHY A BLOOD TEST TELLS YOU ABOUT A COMPARTMENT YOU DID NOT SAMPLE

A lab reports plasma sodium and potassium. Plasma is part of the ECF, so those values describe the ECF well. They describe the ICF only by inference, which is why a normal serum potassium does not guarantee a normal total body potassium.

Two things called a membrane

- **The plasma membrane** is the lipid bilayer around one cell.
- **A tissue membrane** is a sheet of cells lining a cavity, like the pleura or the peritoneum.
- Same word, two scales. The one that matters here is the first.

Lipids build the wall by themselves

1. A phospholipid has a phosphate head that likes water and two fatty acid tails that do not.
2. Drop them in water and they arrange themselves so the tails hide from it. No energy is spent and nothing assembles them.
3. The result is a bilayer: heads out to the water on both sides, tails in the middle.
4. That oily middle is the barrier. It is why a charged particle cannot cross without a protein, which is the whole of the next chapter.

WHAT IS IN THE MEMBRANE, AND WHAT EACH PART DOES

Component	What it contributes
Phospholipids	The bilayer itself, and the oily barrier in the middle
Cholesterol	Buffers fluidity. Keeps the membrane from getting too stiff when cold or too loose when warm
Integral proteins	Cross the membrane. Channels, carriers, pumps and receptors are all integral
Peripheral proteins	Sit on one face. Often anchoring or enzymatic
Glycolipids and glycoproteins	Carbohydrate on the outer face. Cell identity and recognition

FLUID MOSAIC, IN PLAIN TERMS

Fluid because the components drift about in the plane of the membrane rather than being fixed. Mosaic because proteins are scattered through it rather than forming a layer.

Inside the cell

THE ORGANELLES, GROUPED BY WHAT THEY ARE FOR

Organelle	What it does	Worth knowing
Nucleus	Holds DNA, controls transcription	The nuclear envelope has pores that let RNA out
Ribosomes	Build proteins	Free in the cytosol, or bound to rough ER
Rough ER	Makes proteins for export or for membranes	Rough because ribosomes stud it
Smooth ER	Makes lipids, stores calcium	In muscle it is the sarcoplasmic reticulum, the calcium store
Golgi apparatus	Modifies, sorts and packages proteins into vesicles	The despatch department
Lysosomes	Digest what the cell takes in or wears out	Where a phagosome ends up
Peroxisomes	Break down fatty acids and detoxify	Produce and then destroy hydrogen peroxide
Mitochondria	Make ATP	Their own DNA, inherited maternally

The cytoskeleton

THREE FILAMENT TYPES, THINNEST TO THICKEST

Filament	Made of	What it does
Microfilaments	Actin	Shape, movement, the pseudopods of phagocytosis, and contraction with myosin
Intermediate filaments	Various proteins, keratin among them	Mechanical strength. What desmosomes anchor to
Microtubules	Tubulin	Tracks for vesicle transport, the mitotic spindle, and the core of cilia and flagella

From cells to tissues: junctions

A sheet of cells is only useful if the cells are held together and the gaps between them are controlled. Three junction types do three different jobs, and mixing them up is the commonest error in this chapter.

ASK WHAT JOB THE TISSUE NEEDS DONE

What does this junction have to achieve?

Seal the gap between cells

- **Tight junction** (zonula occludens)
- Built from claudins and occludins
- Controls the **paracellular** route
- Intestine, kidney tubule, blood brain barrier

Hold cells together against pulling

- **Desmosome** (macula adherens)
- Cadherins anchored to intermediate filaments
- Resists shear and stretch
- Epidermis, cardiac muscle

Let signals pass cell to cell

- **Gap junction**
- Connexons forming a pore
- Ions and small molecules pass directly
- Cardiac muscle, smooth muscle

Anchor a cell to what is under it

- **Hemidesmosome**
- Anchors a basal cell to the basement membrane
- Keeps an epithelium attached to its base
- The base of the epidermis

THE DISTINCTION WORTH CARRYING INTO THE DISCUSSION

A junction that holds cells together does not necessarily seal the space between them. Desmosomes are strong and leaky; tight junctions are sealing and comparatively weak. A tissue usually needs both.

WHAT HAPPENS WHEN A JUNCTION FAILS

Junction	Failure	What a person notices
Tight junction	The paracellular seal is lost	Leak across the epithelium. In the gut, diarrhoea
Desmosome	Cells come apart under stress	Blistering. Pemphigus vulgaris attacks a desmosomal protein
Gap junction	Cells stop coupling electrically	In the heart, conduction slows and arrhythmia becomes likely

Epithelia: the sheets that decide what enters

- An epithelium is **polarized**: an apical face toward the lumen, a basolateral face toward the blood, and different proteins on each.
- It sits on a **basement membrane**.
- It has no blood vessels of its own, so it is fed by diffusion from below. That limits how thick it can be.
- It is the interface where the transport chapter happens.

NAMED BY SHAPE AND BY LAYERS

	Types	Where and why
Shape	Squamous, cuboidal, columnar	Flat for diffusion, taller where secretion or absorption is the job
Layers	Simple (one), stratified (many), pseudostratified	One layer for exchange, many for protection
Simple squamous	One flat layer	Alveoli and capillaries. As thin as possible, because Fick's law rewards it
Simple columnar	One tall layer	Intestine. Room for the machinery of absorption
Stratified squamous	Many flat layers	Skin and oesophagus. Built to be abraded and replaced

Connective tissue, muscle and nerve, in brief

THE OTHER THREE TISSUE TYPES

Tissue	Defining feature	Examples
Connective	Cells scattered in an extracellular matrix. The matrix is the tissue	Bone, cartilage, blood, fat, tendon
Muscle	Contracts	Skeletal, cardiac, smooth
Nervous	Conducts electrical signals	Neurons and neuroglia

THE SKIN IS ALL FOUR IN ONE ORGAN

Stratified squamous epithelium on top, connective tissue beneath it, smooth muscle at the hair follicles, and nerve endings throughout. If you can find all four in a section of skin, you can find them anywhere.

WORKED PROBLEM 1

A 70 kg adult is 60 percent water. Divide that water into compartments, then say how much blood the patient has if blood is 40 percent cells by volume.

ICF is two thirds of total body water, ECF one third. Plasma is about a quarter of the ECF. One kilogram of water is one litre.

THE WORK

1. Total body water: $70 \times 0.60 = 42 \text{ L}$.
2. Intracellular: $42 \times 2/3 = 28 \text{ L}$. Most of the body's water is inside cells.
3. Extracellular: $42 - 28 = 14 \text{ L}$.
4. Plasma: $14 \times 1/4 = 3.5 \text{ L}$, usually rounded to about 3 L. Interstitial fluid is the rest, about 10.5 L.
5. Blood is plasma plus cells. If cells are 40 percent, plasma is 60 percent, so blood = $3 \div 0.60 = 5 \text{ L}$.
6. **Draw:** a 42 L tank, two thirds inside cells, and of the outside third only a quarter in the vessels.
7. **Carry:** blood is the smallest compartment and the only one we sample easily.

WORKED PROBLEM 2

A patient has watery diarrhoea for two days. Using the compartments, say which one is depleted first, what the body does about it, and why blood pressure is the thing that eventually falls.

Fluid lost from the gut comes out of the ECF. Water moves between compartments by osmosis.

THE WORK

1. Fluid is lost from the gut lumen, which draws directly on the **extracellular** fluid.
2. Plasma is part of the ECF, so plasma volume falls with it.
3. If the fluid lost is roughly isotonic, ECF osmolarity barely changes, so water does not shift out of cells to help. The ECF absorbs the whole loss.
4. Plasma is only about a fifth of the ECF, and the ECF is only a third of body water, so a loss that is small next to total body water is large next to plasma.
5. Falling plasma volume means less venous return, less stroke volume, and eventually a fall in blood pressure.
6. **Draw:** the three compartments with an arrow leaving the ECF, and mark which one the blood pressure sensors are sitting in.
7. **Carry:** the compartment a loss comes out of decides how fast it becomes dangerous, not the size of the loss alone.

PART 2

Transport Across the Membrane

CELLULAR PHYSIOLOGY

What crosses the membrane and how. Diffusion and osmosis, the carriers and the pumps, endocytosis and exocytosis, and how a sheet of cells moves something from the gut into the blood.

- BY THE END**
1. Put any substance on the transport map by asking which way it moves relative to its gradient and who pays.
 2. State the variables in Fick's law and predict the effect of changing each one.
 3. Separate osmolarity from tonicity and predict cell volume in a named solution.
 4. Contrast simple diffusion, facilitated diffusion, primary active and secondary active transport.
 5. Describe the forms of vesicular transport and say which cells use each.
 6. Trace glucose across an intestinal epithelium, naming the protein on each membrane.

The two questions that classify everything

Every transport mechanism in the body answers the same two questions. Ask them in this order and you never have to memorise a list.

THE MAP TO PUT ANY NEW SUBSTANCE ON

Which way is it going relative to its own gradient, and is the cell paying?

Down the gradient, no energy

- **Simple diffusion.** Crosses the lipid unaided
- **Facilitated diffusion.** Needs a protein, still free
- **Osmosis.** The same thing, for water

Against the gradient, energy spent

- **Primary active.** The protein burns ATP itself
- **Secondary active.** Spends a gradient another pump built

Too big for any protein

- **Endocytosis.** Wrapped in membrane, brought in
- **Exocytosis.** Wrapped in membrane, sent out

Across a whole sheet of cells

- **Transcellular.** Through the cells, two membranes
- **Paracellular.** Between them, past the tight junctions

HOLD ONTO THIS

A student who knows the sodium potassium pump is primary and SGLT1 is secondary has understood the only difference that matters: where the ATP was spent.

Simple diffusion, and what sets its speed

What diffusion is

- Molecules move from where there are more of them to where there are fewer.
- Nobody pushes them. There are simply more ways to be spread out than stacked in a corner.

- No energy and no protein required.
- At equilibrium molecules keep moving both ways, but the two rates are equal. Equilibrium is a standoff, not a stillness.

Fick's law, the four variables

CHANGE ONE AND THE RATE FOLLOWS

Variable	Effect on rate	What it looks like clinically
Concentration gradient	Steeper gradient, faster	A low arterial oxygen widens the alveolar gradient
Surface area	More area, faster	Emphysema destroys alveolar walls and cuts area
Membrane thickness	Thicker, slower	Fibrosis thickens the barrier
Diffusion distance	Further, slower	Pulmonary oedema puts fluid in the way

What crosses the lipid unaided, in order

1. **Small nonpolar gases.** Oxygen, carbon dioxide. Straight through, fast.
2. **Small uncharged polar molecules.** Water, urea. Slowly.
3. **Large polar molecules.** Glucose, amino acids. Not without a protein.
4. **Ions.** Sodium, potassium, calcium, chloride. Never without a protein, whatever their size.

WHY THIS ORDER IS WORTH KNOWING COLD

It decides which mechanism a substance is forced to use. If a question tells you the molecule is charged, you already know a protein is involved before you know anything else about it.

WORKED PROBLEM 1

A patient develops pulmonary oedema, which puts a layer of fluid between the air and the blood and doubles the distance oxygen must travel. Nothing else changes. What happens to oxygen transfer, and why is the patient breathless rather than simply tired?

Fick's law: rate rises with gradient and area, falls with thickness and distance.

THE WORK

1. Distance went up. Gradient, area and thickness are unchanged.
2. Distance sits in the denominator, so doubling it roughly **halves** the rate of transfer.
3. Less oxygen crosses per second, so arterial oxygen falls.
4. Chemoreceptors respond to that fall by driving ventilation up, and that is what the patient feels as breathlessness.
5. **Draw:** two boxes with a barrier between, the barrier drawn twice, thin and thickened, distance labelled on both.
6. **Carry:** nothing was wrong with the oxygen, the haemoglobin or the gradient. One geometric variable moved and the system followed.

Osmosis, osmolarity and tonicity

Osmosis in one line

- Water moves toward the side with **more solute**, because more solute means less water per litre.
- Track the water, not the solute. Tracking the solute is where marks are lost.
- Osmotic pressure is the pressure needed to **stop** that movement, not a push the solutes exert.

The two words students confuse

OSMOLARITY AGAINST TONICITY

	Osmolarity	Tonicity
What it counts	Every dissolved particle	Only solutes that cannot cross the membrane
Depends on a membrane?	No. Measure it in a beaker	Yes. The word is meaningless until you name a membrane
Normal body value	About 300 mOsM	Isotonic is about 300 mOsM of nonpenetrating solute
Tells you cell volume?	No	Yes. This is the one that matters to a cell

THE SINGLE MOST TESTED IDEA IN THIS HALF

A 300 mOsM urea solution is **isosmotic** with the cell and will still **burst** a red blood cell. Urea crosses the membrane, enters the cell, and water follows it in. Isosmotic and hypotonic at the same time. The two words are not synonyms.

WORKED PROBLEM 2

A red cell is dropped into each of three solutions. A: 300 mOsM sodium chloride. B: 300 mOsM urea. C: 150 mOsM sodium chloride. Sodium chloride does not cross the red cell membrane; urea does. Predict each, and give both words.

Cell interior is about 300 mOsM, all of it nonpenetrating.

THE WORK

1. **A.** Nonpenetrating outside equals nonpenetrating inside. Water has no reason to move. **Isosmotic, isotonic, no volume change.**
2. **B.** Count only nonpenetrating solute outside: zero. The cell has 300 inside and there is none outside, so water moves in.
3. Urea also moves in down its own gradient, raising inside osmolarity and pulling in still more water. The cell swells and lyses. **Isosmotic but hypotonic.**
4. **C.** Half the nonpenetrating solute outside that the cell has inside, so water moves in and the cell swells. **Hyposmotic and hypotonic.**
5. **Draw:** three cells in three beakers, a water arrow in each, and write both words separately under each so you are forced to notice when they disagree.
6. **Carry:** osmolarity is a property of a solution. Tonicity only exists once a membrane is named.

Facilitated diffusion

- Protein mediated, still **passive**, still downhill, still no ATP. The protein is a door, not an engine.
- Take the gradient away and movement stops. Reverse the gradient and the protein carries the cargo the other way.
- Glucose enters most cells this way, through the GLUT family.

TWO FEATURES THAT COME WITH USING A PROTEIN

Feature	What it means	How you spot it on a graph
Specificity	A carrier binds a particular shape. GLUT carries sugars, not amino acids	Not visible on a rate graph; visible in what the cell will and will not take up
Saturation	A finite number of carriers. Fill them all and the rate stops rising	The curve flattens. Simple diffusion is a straight line that keeps climbing

Channels against carriers

BOTH ARE PROTEINS, AND THEY WORK DIFFERENTLY

	Channel	Carrier
Shape	An open pore	A binding site that changes shape
Speed	Very fast, millions of ions a second	Much slower
Cargo	Ions and water	Larger solutes such as glucose
Controlled by	Gating: voltage, chemical or mechanical	Availability and saturation

Primary active transport

A cell at equilibrium with its surroundings is dead. Life is the business of holding gradients that would rather collapse, and this is the protein that does it.

THE SODIUM POTASSIUM ATPASE, ONE CYCLE

- 1 Three sodium ions bind inside**
The pump is open to the cytosol and sodium is scarce there, so it binds readily.
- 2 ATP is hydrolysed**
A phosphate is transferred to the pump and it changes shape.
- 3 Three sodium are released outside**
Against their gradient, into a fluid already high in sodium.
- 4 Two potassium ions bind outside**
The new shape has a high affinity for potassium.
- 5 The phosphate comes off**
The pump returns to its original shape.
- 6 Two potassium are released inside**
Against their gradient. The cycle is ready to run again.

WHAT THE PUMP BUYS THE CELL

Consequence	Why it follows	Where it matters later
Low sodium inside	Three sodium leave per cycle, continuously	Every secondary transporter runs on this gradient
High potassium inside	Two potassium enter per cycle	Sets up the resting membrane potential in Week 3
The cell is electrogenic	Three positive out, two positive in, net one positive out	A small direct contribution to the negative resting potential
A large metabolic bill	A substantial share of resting metabolic rate	Explains why ischaemia causes cells to swell within minutes

Other primary pumps worth naming

- **Calcium ATPase.** Holds cytosolic calcium about ten thousand times lower than outside, which is what makes calcium usable as a signal.
- **Proton pump.** Acidifies gastric juice, and the target of proton pump inhibitors.

Secondary active transport

- Moves one substance **down** its gradient and uses the energy released to drag a second **up** its own.
- **No ATP touches this protein.** The ATP was spent earlier by the sodium potassium pump, building the gradient this one spends.
- Poison the pump and every secondary transporter in the cell fails shortly after.

THE TWO SHAPES, AND THE NAMES ARE LITERAL

Type	Direction	Example	What it achieves
Symport	Both the same way	SGLT1: sodium in, glucose in	Glucose absorbed against its gradient in gut and kidney
Antiport	Opposite ways	Sodium calcium exchanger: sodium in, calcium out	Keeps cytosolic calcium low between beats

DIGOXIN, FOUR STEPS AND NO NEW PHYSIOLOGY

- 1 Digoxin blocks the sodium potassium pump**
Sodium stops leaving the cell.
- 2 Intracellular sodium rises**
The inward sodium gradient gets weaker.
- 3 The sodium calcium exchanger slows**
It runs on that gradient, so it exports less calcium.
- 4 Intracellular calcium rises**
More calcium available per beat, so the heart contracts more forcefully.

WORKED PROBLEM 3

A patient with cholera is losing salt and water fast and no intravenous fluid is available. Oral rehydration solution, water with salt and glucose in it, saves them. Plain salt water does not work nearly as well. Explain why the glucose matters.

Cholera toxin drives chloride and water secretion into the lumen. It does not damage SGLT1 on the apical membrane.

THE WORK

1. SGLT1 is a **symport**: sodium in down its gradient, glucose in against its own, the sodium paying for the glucose.
2. It needs sodium in the lumen and a sodium gradient across the apical membrane. The basolateral pump keeps intracellular sodium low, so the gradient is there.
3. With glucose present, every glucose absorbed brings a sodium in with it. Salt alone gives no such coupling.
4. Sodium accumulating inside is pumped out basolaterally, raising osmolarity in the interstitial fluid.
5. Water follows that solute by osmosis, out of the lumen and into the body. Water is never actively transported; it follows the sodium.
6. **Draw**: one intestinal cell, apical and basolateral membranes, SGLT1 apical, the pump basolateral, a dotted water arrow following the sodium.
7. **Carry**: the treatment works because the toxin breaks one pathway and leaves another intact. Knowing which protein sits on which membrane is what lets you see that.

Transport maximum

- Any protein mediated transport can saturate, because the number of proteins is finite.

- **Below the T_m :** raise the concentration, raise the rate.
- **At the T_m :** raise the concentration, nothing happens. The excess goes wherever it was headed.
- **Simple diffusion never saturates.** A graph that flattens means a protein is involved.

WHY UNCONTROLLED DIABETES PUTS SUGAR IN THE URINE

- 1 Plasma glucose rises**
More glucose is filtered at the glomerulus.
- 2 Filtered load passes the renal threshold**
About 180 to 200 mg/dL of plasma glucose in most people.
- 3 The reabsorption carriers saturate**
The transport maximum is around 375 mg/min.
- 4 Glucose remains in the tubule**
It appears in the urine. That is glycosuria.
- 5 Water is held in the tubule with it**
Glucose is osmotically active, so urine volume rises and the patient is thirsty.

ONE SATURATED CARRIER, THREE SYMPTOMS

Sugar in the urine, high urine volume and thirst all come from the same ceiling being crossed. Learn the ceiling and the three fall out of it.

Vesicular transport

Proteins, bacteria and droplets of fluid are far too large for any channel or carrier, so the cell wraps them in a piece of its own membrane.

THE FIVE, AND HOW THEY DIFFER

Process	Direction	How selective	Who does it
Phagocytosis	In	A large particle: a bacterium, a dead cell	Specialised cells only: macrophages, neutrophils
Pinocytosis	In	Nonselective, whatever is dissolved nearby	Most cells, continuously
Receptor mediated endocytosis	In	Very selective, a specific ligand at a receptor	Most cells. LDL and transferrin uptake
Exocytosis	Out	The vesicle contents, plus its membrane proteins	All cells. Regulated release is triggered by calcium
Transcytosis	In one face, out the other	Carries cargo straight across a cell	Capillary endothelium, for large proteins

RECEPTOR MEDIATED ENDOCYTOSIS, STEP BY STEP

- 1 The ligand binds its surface receptor**
LDL binding the LDL receptor, for example.
- 2 Loaded receptors gather in a coated pit**
The inner face of that patch is coated with clathrin.
- 3 The pit invaginates and pinches off**
A coated vesicle enters the cytosol.
- 4 The coat is shed and the vesicle sorts its cargo**
Receptors are usually recycled to the surface.
- 5 The cargo is delivered**
LDL goes to the lysosome and its cholesterol is released for use.

WORKED PROBLEM 4

A child has a plasma cholesterol several times normal from a young age, with the same pattern in the family. The LDL receptor is defective. Explain why plasma cholesterol is high, and name the step that failed.

LDL carries cholesterol in blood. Cells take it up by receptor mediated endocytosis.

THE WORK

1. LDL is far too large to cross by any protein route, so uptake must be **receptor mediated endocytosis**.
2. The receptor binds LDL at the surface and loaded receptors gather in a clathrin coated pit.
3. With a defective receptor, LDL is never bound, so no pit forms and no vesicle is taken in. **The very first step fails.**
4. The cholesterol is not missing from the body. It is in the plasma and cannot get out of it.
5. Cells sense they are short of cholesterol and synthesise more of their own, which makes plasma levels worse.
6. **Draw:** a cell surface with receptors and LDL outside, a coated pit forming in the normal case, then the same drawing with the receptor crossed out.
7. **Carry:** a transport defect does not always mean too little of something. Here there is far too much, in the wrong compartment, because the door into the right one is broken.

Transepithelial transport

Everything above crossed one membrane. Absorbing a nutrient means crossing a whole sheet of cells, in one side and out the other.

- An epithelial cell is **polarized**: the apical membrane facing the lumen carries different proteins from the basolateral membrane facing the blood.
- **Tight junctions** are what keep the two sets from mixing. This is where the anatomy of the first half and the physiology of this half meet.
- There are two routes across: **transcellular**, through the cells, and **paracellular**, between them.

GLUCOSE FROM THE GUT INTO THE BLOOD, THREE PROTEINS ON TWO MEMBRANES

- 1 SGLT1, apical membrane**
Secondary active transport. Sodium comes in down its gradient and drags glucose in against its own.
- 2 GLUT2, basolateral membrane**
Facilitated diffusion. Glucose leaves down its gradient into the interstitial fluid.
- 3 Sodium potassium ATPase, basolateral membrane**
Primary active transport. Throws sodium out so the apical gradient never runs down.
- 4 Water follows**
By osmosis, transcellular and paracellular, following the solute that just moved.

THREE MECHANISMS, ONE MOLECULE, ONE TRIP

This is the pathway to draw from blank paper. At every step ask the same two questions: which way relative to the gradient, and who is paying.

HOW LEAKY THE PARACELLULAR ROUTE IS DEPENDS ON THE JUNCTIONS

Epithelium	Tight junctions	What that allows
Proximal tubule	Leaky	A good deal passes between the cells. Bulk reabsorption
Distal nephron	Tight	Almost nothing passes between. Fine control of what is kept
Bladder	Very tight	Urine is held without leaking back into the body
Small intestine	Moderately leaky	Water and some ions follow solute between the cells

WORKED PROBLEM 5

A drug blocks only the basolateral sodium potassium pump in an intestinal cell. Glucose absorption across the epithelium falls to almost nothing within minutes, even though the drug never touched SGLT1. Explain the sequence.

SGLT1 is apical and is a sodium glucose symport. GLUT2 is basolateral and is facilitated diffusion.

THE WORK

1. The pump was throwing sodium out basolaterally, holding intracellular sodium low.
2. Block it and sodium stops leaving. Intracellular sodium begins to rise.
3. SGLT1 runs on the sodium gradient across the apical membrane. As intracellular sodium rises, that gradient shrinks.
4. With a small gradient there is little energy to spend, so SGLT1 can no longer drag glucose in against its gradient. Apical uptake falls.
5. Nothing is wrong with SGLT1. Its fuel supply was cut off two steps upstream.
6. **Draw:** the polarized cell again, the pump crossed out, rising sodium written inside, the arrow through SGLT1 getting thinner.
7. **Carry:** this is what secondary means. Every secondary transporter is one pump failure away from stopping.

PART 3

How Cells Talk to Each Other

CELL SIGNALING

One signal molecule reaching the right cell, being recognised, converted into something the inside of the cell can act on, amplified enough to matter, and then switched off.

- BY THE END**
1. Name the signal types by how far they travel and say what sets the speed and reach of each.
 2. Predict where a receptor sits, and the onset and duration of the response, from the solubility of the signal.
 3. Draw the G protein cascade from blank paper, including how it switches itself off.
 4. Name the second messengers and the amplifier enzyme that makes each one.
 5. Explain amplification and say why a cascade with high gain must terminate precisely.
 6. Tell an agonist from an antagonist, and a competitive blocker from a noncompetitive one, from a dose response curve.

Every pathway is the same five steps

LEARN THE SHAPE ONCE AND THE REST OF THE COURSE IS EXAMPLES

- 1 Signal**
 A molecule is released by one cell.
- 2 Receptor**
 A protein binds it, and binding changes the receptor's shape. That shape change is how information gets in without the molecule entering.
- 3 Transduction**
 The message is converted into something the inside of the cell can act on.
- 4 Response**
 The cell opens a channel, runs an enzyme, secretes, contracts or divides.
- 5 Termination**
 The signal is switched off. Without this step the pathway is a disease, not a control system.

Signal types, and how far they travel

SORTED BY DISTANCE, WHICH SETS SPEED AND REACH

Type	How it travels	Distance	Speed and reach
Gap junctional	Directly cell to cell through connexon pores	Touching cells	Fastest. Nothing enters the extracellular fluid at all
Contact dependent	A membrane molecule binds a receptor on the cell it touches	Touching cells	Immune cells checking each other
Autocrine	Released and acts on the cell that released it	Itself	Common in immune signalling and growth control
Paracrine	Into the interstitial fluid, onto neighbors	Short. Broken down before it travels	Histamine in an allergic response
Neurotransmitter	Across a synaptic cleft onto one target	Very short	Very fast, very precise
Hormone	Into the blood	Everywhere	Slow to arrive, long lasting, reaches every cell

NO RECEPTOR, NO RESPONSE

A hormone in the blood touches every cell you have. What decides which cells respond is not where the signal goes, it is which cells carry a receptor for it. That single idea explains how one hormone does different things in different tissues and nothing at all in most of them.

Where the receptor sits, and what decides that

There are only two places a receptor can be, and one property of the signal decides which.

SOLUBILITY DECIDES EVERYTHING DOWNSTREAM

Can the signal molecule cross a lipid bilayer?

Lipophobic, water soluble

- Peptides, proteins, catecholamines such as epinephrine
- **Cannot** cross the membrane
- Receptor is **on the surface**
- Relayed by proteins already built and waiting
- Onset in **seconds**, stops soon after the signal leaves

Lipophilic, lipid soluble

- Steroid hormones, thyroid hormone
- **Passes straight through** the membrane
- Receptor is **inside**, cytosol or nucleus
- Acts on DNA, changes which proteins are made
- Onset in **hours**, lasts hours to days after the signal has gone

YOU ARE NOT MEMORISING ONSET AND DURATION

You are working them out. Steroid means lipophilic, means intracellular receptor, means gene transcription, means slow and long. Every step follows from the one before it.

One thing to be careful about

- Lipophilic hormones do not dissolve well in blood, so they travel **bound to carrier proteins**.
- Only the **free** fraction can enter cells.
- That is why a total hormone level and a free hormone level are different measurements, and why the free one is usually the one that matters.

WORKED PROBLEM 1

Patient A is given a peptide hormone and reports an effect within minutes that has worn off an hour later. Patient B is given a steroid and feels nothing for hours, but the effect is still measurable two days later. Explain both time courses from the chemistry alone.

Peptides are lipophobic. Steroids are lipophilic.

THE WORK

1. **A.** A peptide cannot cross the membrane, so its receptor is on the surface.
2. Surface receptors work through proteins already in the cell: G proteins, enzymes, second messengers. Nothing has to be built, so the response begins in seconds to minutes.
3. Those second messengers are broken down continuously, so when the hormone leaves the signal collapses quickly. Short onset, short duration.
4. **B.** A steroid crosses the membrane and binds a receptor inside the cell.
5. The complex acts on DNA and changes which proteins the cell makes. Transcription and translation take hours; that is the delay.
6. Once made, those proteins persist until they are degraded, so the effect outlives the hormone. Slow onset, long duration.
7. **Draw:** two cells side by side, one with the signal stopping at the surface and a cascade inside, one with the signal walking through to the nucleus. Put a clock on each.
8. **Carry:** onset and duration are not facts per hormone. They fall out of where the receptor is, which falls out of solubility.

The four families of surface receptor

SORT ALMOST ANYTHING INTO ONE OF THESE

Family	How it works	Speed	Example
Receptor channel	The receptor is itself an ion channel. Ligand binds, channel opens	Milliseconds, the fastest	The neuromuscular junction
G protein coupled	Receptor activates a G protein, then an amplifier enzyme, then a second messenger	Seconds	Epinephrine at a beta receptor
Receptor enzyme	An enzyme site, usually a kinase, on the receptor's inner face	Seconds to minutes	The insulin receptor
Integrin	Linked to the cytoskeleton, transduces mechanical information	Varies	Sensing what the cell is attached to

The G protein cascade, step by step

This is the pathway to draw from blank paper, because so much else is a variation on it. The parts: a receptor crossing the membrane seven times, a G protein with alpha, beta and gamma subunits, an amplifier enzyme, and a second messenger.

ONE PASS THROUGH THE CASCADE, AND HOW IT SWITCHES ITSELF OFF

- 1 At rest**
The alpha subunit holds GDP. Everything is quiet.
- 2 Ligand binds the receptor outside**
The receptor changes shape.
- 3 The receptor grips the G protein**
The alpha subunit swaps GDP for GTP. This is the switch turning on.
- 4 The alpha subunit separates**
It moves along the inner face of the membrane to the amplifier enzyme.
- 5 The amplifier enzyme runs**
It produces a great deal of second messenger.
- 6 A protein kinase is activated**
It adds phosphate groups to target proteins and changes what they do. Phosphorylation is the cell's general purpose on switch.
- 7 The alpha subunit hydrolyses its own GTP**
It reassembles with beta and gamma. The switch turns itself off.

THE G PROTEIN IS A TIMER AS WELL AS A SWITCH

How long the alpha subunit takes to hydrolyse its GTP sets how long the signal lasts. Cholera toxin locks that subunit so it cannot, the switch jams on, cAMP stays high, chloride and water pour into the gut. That is the diarrhoea.

Second messengers

The hormone is the first messenger and never gets in. The second messenger is what carries the news around the cytosol.

THE ONES TO KNOW

Messenger	Amplifier enzyme	What it activates	Seen in
Cyclic AMP	Adenylyl cyclase, from ATP	Protein kinase A	Epinephrine at a beta receptor
IP3 and DAG	Phospholipase C, cutting a membrane phospholipid in two	IP3 opens calcium channels on the ER; DAG activates protein kinase C	Many peptide hormones
Calcium	Released from the ER or entering from outside	Calmodulin, troponin	Contraction, secretion, almost everything
Cyclic GMP	Guanylyl cyclase	Protein kinase G	Nitric oxide relaxing smooth muscle

WHY CALCIUM WORKS AS A MESSENGER AT ALL

The cell holds cytosolic calcium about ten thousand times lower than outside, using the calcium pumps from the transport chapter. Opening a channel for a moment therefore produces a large proportional change. The low resting level is what makes the signal readable.

Amplification

- A hormone is present in blood at roughly a billionth of a mole per litre. The response can be enormous.
- The gap between those two is amplification, and it is a property of the **cascade**, not of the hormone.

WHERE THE GAIN COMES FROM, EPINEPHRINE IN A LIVER CELL

- 1 One ligand, one receptor**
But the receptor activates many G proteins while the ligand is bound.
- 2 Each G protein switches on an amplifier enzyme**
Multiplication one.
- 3 Each enzyme makes many molecules of second messenger**
Multiplication two, and the biggest.
- 4 Each second messenger activates a kinase, which phosphorylates many targets**
Multiplication three.
- 5 Each phosphorylated enzyme works on many glycogen molecules**
Multiplication four. One hormone molecule ends up producing on the order of a hundred million glucose molecules.

TWO CONSEQUENCES, AND THEY PULL IN OPPOSITE DIRECTIONS

Consequence	Why	What it forces
Sensitivity	Four multiplying steps	A cell can respond to a vanishingly small signal, which is how hormones work at the concentrations they do
Fragility	The same four steps	A cascade with that much gain must be shut off precisely, or a trace of signal runs away

Receptor enzymes and intracellular receptors

THE INSULIN RECEPTOR, AND WHY IT TIES THIS WEEK TOGETHER

- 1 Insulin binds its receptor**
No G protein. The receptor itself carries a tyrosine kinase site on its inner face.
- 2 The receptor autophosphorylates**
Then phosphorylates a chain of intracellular proteins in turn.
- 3 Vesicles carrying GLUT4 move to the surface**
And fuse with the plasma membrane.
- 4 Glucose uptake rises in muscle and fat**
The last step is exocytosis inserting a facilitated diffusion carrier: three ideas from this week in one pathway.

Intracellular receptors, the lipophilic route

1. The hormone crosses the membrane on its own.
2. It binds its receptor in the cytosol or the nucleus.
3. The complex binds DNA at a response element.
4. Transcription of specific genes changes, so the cell makes different proteins.
5. The cell's behavior changes for as long as those proteins last.

Cortisol and the sex steroids work this way. So does thyroid hormone, which is not a steroid but is lipophilic and behaves like one here.

Turning the volume up and down

THE CELL CHANGES HOW LOUDLY IT HEARS

	What happens	When	Where you meet it
Down regulation	Receptors removed from the surface or fewer made	Sustained high signal	Drug tolerance. Why a constant infusion works less well than a pulsed one
Up regulation	More receptors made, the cell becomes more sensitive	Sustained low signal	A denervated muscle becomes hypersensitive to acetylcholine

MOLECULES THAT BIND RECEPTORS

Term	Binds?	Activates?	What it does to the response
Full agonist	Yes	Yes	Can produce the maximum response
Partial agonist	Yes	Partly	Activates, but cannot reach the maximum however much you give
Competitive antagonist	Yes, same site	No	Blocks, but enough agonist outcompetes it. Curve shifts right, maximum unchanged
Noncompetitive antagonist	Yes, elsewhere	No	Cannot be outcompeted. The maximum itself comes down

WHY THE COMPETITIVE DISTINCTION IS CLINICAL, NOT ACADEMIC

It decides whether more agonist can rescue the patient. You can out dose a competitive blocker. You cannot out dose a noncompetitive one; you support the patient until the drug clears.

Signal termination

Nothing above matters unless the signal can be stopped.

FIVE WAYS A SIGNAL ENDS

Route	How	Example
The ligand leaves or is destroyed	Broken down, taken back up, or cleared	Acetylcholinesterase clears the cleft in about a millisecond
The receptor stops responding	Internalised or chemically modified so it no longer couples	Receptor desensitisation
The switch turns itself off	The G protein hydrolyses its GTP	Built into every G protein cascade
The second messenger is destroyed	Enzymes break it down continuously	Phosphodiesterase breaks down cAMP; calcium is pumped back out
The phosphate comes off	Phosphatases reverse what the kinases did	Every on switch has a matching off switch

FAILURE TO TERMINATE IS A CATEGORY OF DISEASE

What fails	Mechanism	Result
Cholera toxin	Locks the G protein alpha subunit on	cAMP stays high, chloride and water pour into the gut
Some endocrine tumours	A mutation leaves a receptor active with no ligand at all	Continuous hormone output with no signal
Sildenafil, on purpose	Inhibits the phosphodiesterase that breaks down cGMP	The relaxing signal lasts longer. Termination blocked as the therapy

KNOWING HOW A PATHWAY IS SWITCHED OFF TELLS YOU WHERE A DRUG CAN ACT

Half the drugs in this chapter do not touch the receptor at all. They work on the off switch.

Dose and response

- Plot response against dose and the curve rises slowly, then steeply, then flattens.
- The flattening is **saturation**: a finite number of receptors, all occupied. Past that point more drug adds nothing and side effects are all that is left.
- It is the same idea as the transport maximum in the last chapter.
- Plot dose on a **logarithmic** axis and the same data becomes a tidy S shape, which is why nearly every published curve is drawn that way.

TWO NUMBERS, AND THEY ARE INDEPENDENT

	What it measures	What it means	A low or high value
EC50	Potency	The dose giving half the maximum response	Lower EC50 means more potent, you need less drug
Maximal response	Efficacy	How much the drug can do at its best	A higher ceiling means it can do more

THE COMMONEST ERROR HERE

A very potent drug with a low ceiling is worse for a severe problem than a less potent drug with a high one. A partial agonist has lower efficacy no matter how potent it is. Potency is how much you need; efficacy is how much you get.

WORKED PROBLEM 2

Two dose response curves are drawn for the same agonist. In experiment one a blocker shifts the curve right with the same maximum. In experiment two a different blocker lowers the maximum with no rightward shift. Identify each, and say which patient you could rescue with more agonist.

Competitive antagonists bind the same site as the agonist. Noncompetitive ones do not.

THE WORK

1. **One.** The maximum is unchanged, so every receptor can still give a full response if the agonist reaches it.
2. That is a **competitive antagonist**. The two compete for one site, so raising agonist wins the competition.
3. The rightward shift means a higher EC₅₀: more drug for the same effect. Potency fell, efficacy did not.
4. **Two.** The maximum is lower, so some receptors cannot respond at any agonist concentration.
5. That is a **noncompetitive antagonist**. Raising the agonist does not help, because the problem is not competition for the site.
6. Efficacy fell. EC₅₀ may be unchanged.
7. **Clinically:** you can out dose the competitive blocker. You cannot out dose the other one.
8. **Draw:** one set of axes, three curves, control and both blockers, with EC₅₀ marked on each.